

EXPERIMENT 14

IMPLEMENTATION OF ELLIPTIC CURVE CRYPTOGRAPHY(ECC)

AIM

To implement a simple Elliptic Curve Cryptography (ECC) key generation program in C.

ALGORITHM

1. Start the program.
2. Enter the private key.
3. Define the generator point.
4. Generate the public key.
5. Display the private and public keys.
6. Stop the program.

CODE

```
#include <stdio.h>

int main()
{
    int privateKey;
    int Gx = 5, Gy = 9;
    int publicX, publicY;

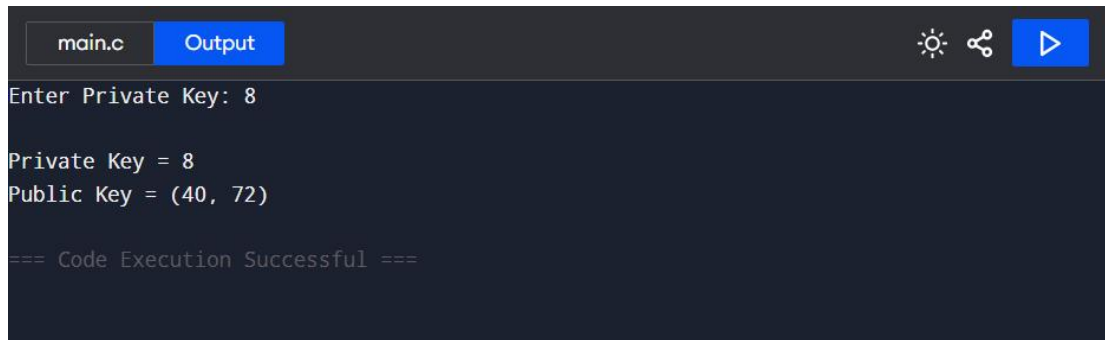
    printf("Enter Private Key: ");
    scanf("%d",&privateKey);

    publicX = privateKey * Gx;
    publicY = privateKey * Gy;

    printf("\nPrivate Key = %d",privateKey);
    printf("\nPublic Key = (%d, %d)",publicX,publicY);

    return 0;
}
```

OUTPUT



The screenshot shows a code editor interface with a dark theme. At the top, there are two tabs: 'main.c' and 'Output'. The 'Output' tab is active, displaying the program's output. The output text is as follows:

```
Enter Private Key: 8  
  
Private Key = 8  
Public Key = (40, 72)  
  
=== Code Execution Successful ===
```

At the top right of the output window, there are three icons: a gear (settings), a share icon, and a blue button with a white play icon.

RESULT

The public key was successfully generated from the private key using a simplified ECC implementation.